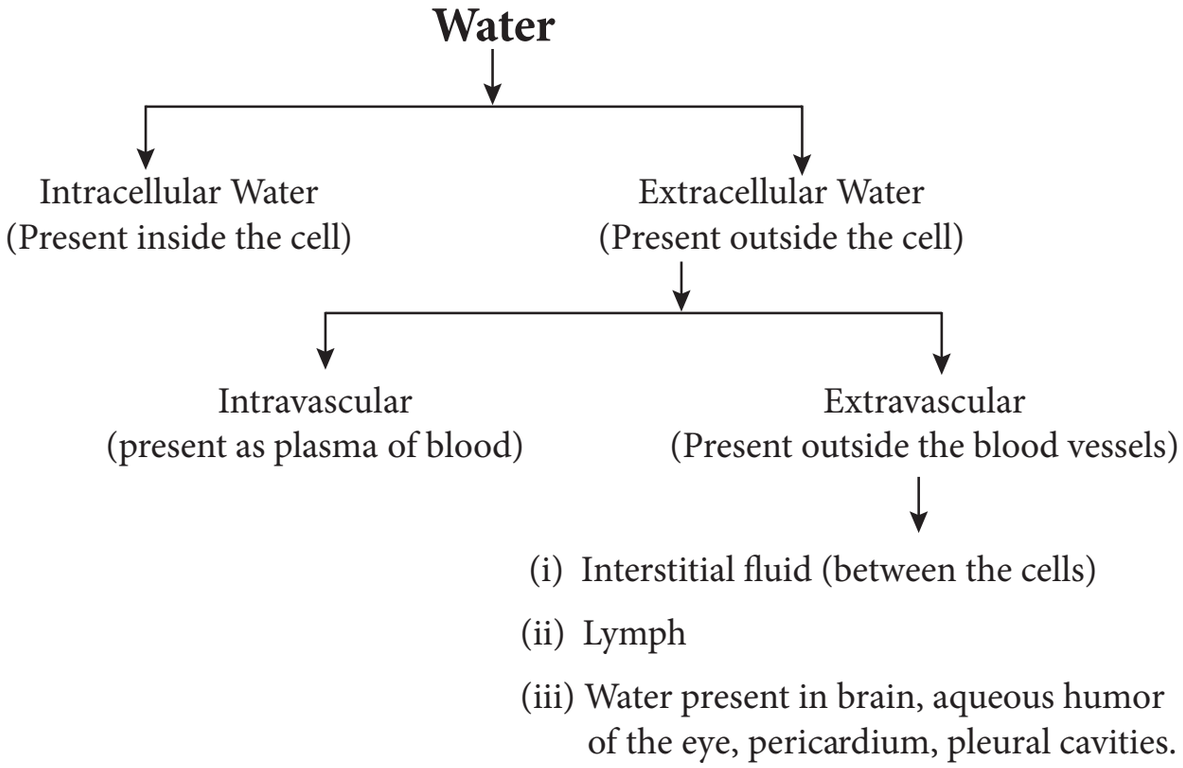


Fig 11.27: Distribution of water in the body

11.5.1 Distribution of water in the body

Total body water content is mainly determined by total amount of salt in the body. Salt and water concentration in the body is controlled by the kidneys.

11.5.2 Water in the body – Intracellular and Extracellular fluid

Water flows in and out of the body cells through cell membranes.

Intracellular fluid:

This is fluid contained within a cell represents about two thirds of all body fluids.

Extracellular fluid:

This is fluid present outside the cells, and represents about one-third of all body fluid.

Extracellular fluid is further divided into interstitial fluid which is water between cells and intravascular fluid which is water in the blood stream and lymph. Interstitial fluid forms a transport link between tissue cells and the blood.

11.5.3 Sources of water

The body has three sources of water. Besides drinking water, the following are the sources of water.

1. The water contained in food, eg. Fruits and vegetables contain 80-90 percent water. Milk contains 80-88 percent, meat contains 40-75 percent, flour, crackers and bread contain 5-35 percent of water.
2. In addition to water ingested fluids such soups and beverages also supply essential minerals and vitamins.
3. Metabolic water is formed by the metabolism of food in the body. It may amount to about 450ml per day.

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